# On the origin of functions by means of natural selection and other processes

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*ABSTRACT*

I at least try to write my paper before I write my abstract, because I hardly ever know what to summarize before I've written it. This in itself is most likey to be seen as a shortcoming, but be that as it may.

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If we are to believe Amie L. Thomasson, then many ontological and metaphysical theories have passed their best-before dates. Her arguments are that current ontologies is far too complex (Thomasson, 2016) and that metaphysics itself is basically broken (Thomasson, 2026), and that it therefore must be made *functional* again. By functional she does not mean that someone should repair metaphysics but that the old quinean ontologies should be retired. In metaphysics entities should be studied based on their functions. Not their properties, which is actually more less the opposite of Quintes view. Quine wrote

> a theory is [ontologically] committed to those and only those entities to which the bound variables of the theory must be capable of referring in order that the affirmations made in the theory be true (Quine, 1948).

Another way to think of this would be to choose ones study objects based on whether they are amenable to first order logic.

Following Quine's seminal paper, philosophers started to interest themselves in metaphysical issues after the low-water mark after the onslaught of the Vienna school and the positivists. During the following renaissance metaphysicians studied explanatory power of entities and concepts, their structures, truthmakers and fundamentality and grounding (Thomasson, 2025).

It would be to go far beyond the scope of this text to review those various metaphysical disciplines; rather we go directly to Thomasson's preferred road forward, which is the functional approach strengthened with conceptual engineering (Thomasson, 2025, p. 103–131). Ruth Garrett Millikan is the pioneer in building a modern theory of functions, which is built on a solid ground. She thinks of functions as things that are somehow designed for a purpose.

> My claim will be that it is the "proper function" of a thing that puts it in a biological category, and this has to do not with its powers but with its history. Having a proper function is a matter of having been "designed to" or of being "supposed to" (impersonal) perform a certain function. The task of the theory of proper functions is to define this sense of "designed to" or "supposed to" in naturalist, nonnormative, and nonmysterious terms (Millikan, 1984, p. 17).

A biological category is given by its proper function. For instance, molluscs have a body fluid called *haemolymph* which belongs to the same biological category as blood (Machałowski and Jesionowski, 2021). Together with a two

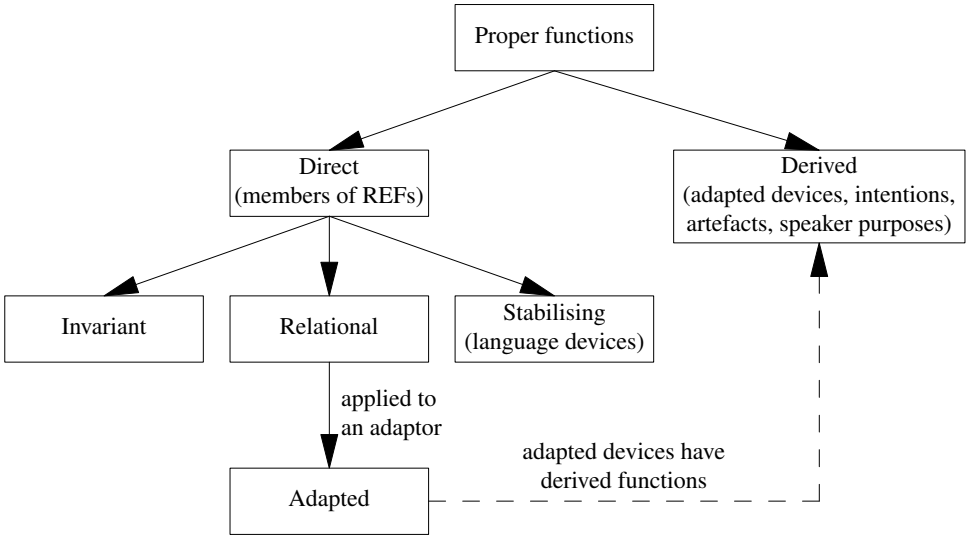


Figure 1. A taxonomy of different forms of functions accoring to (Millikan, 1984). A proper function can be either direct or derived.

chamber heart haemolymph is distributing oxygene and nutrients through the body of any slug, that is its circulatory system.

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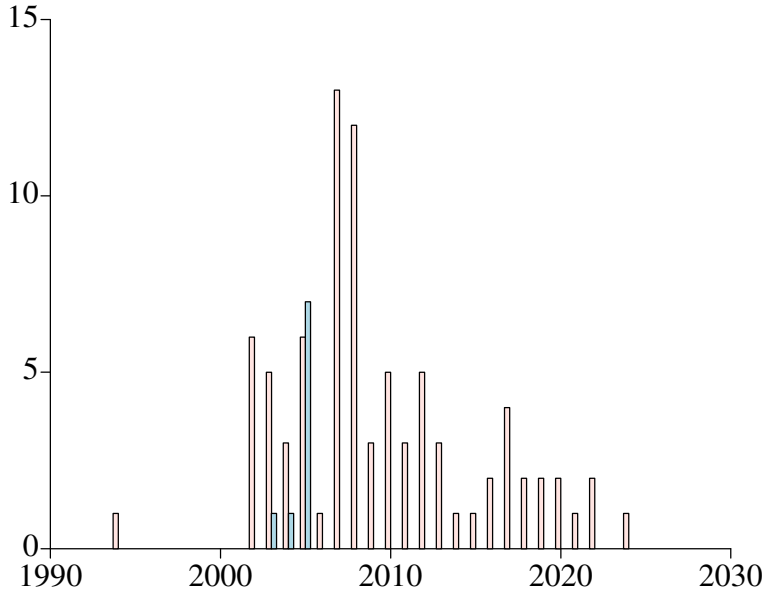
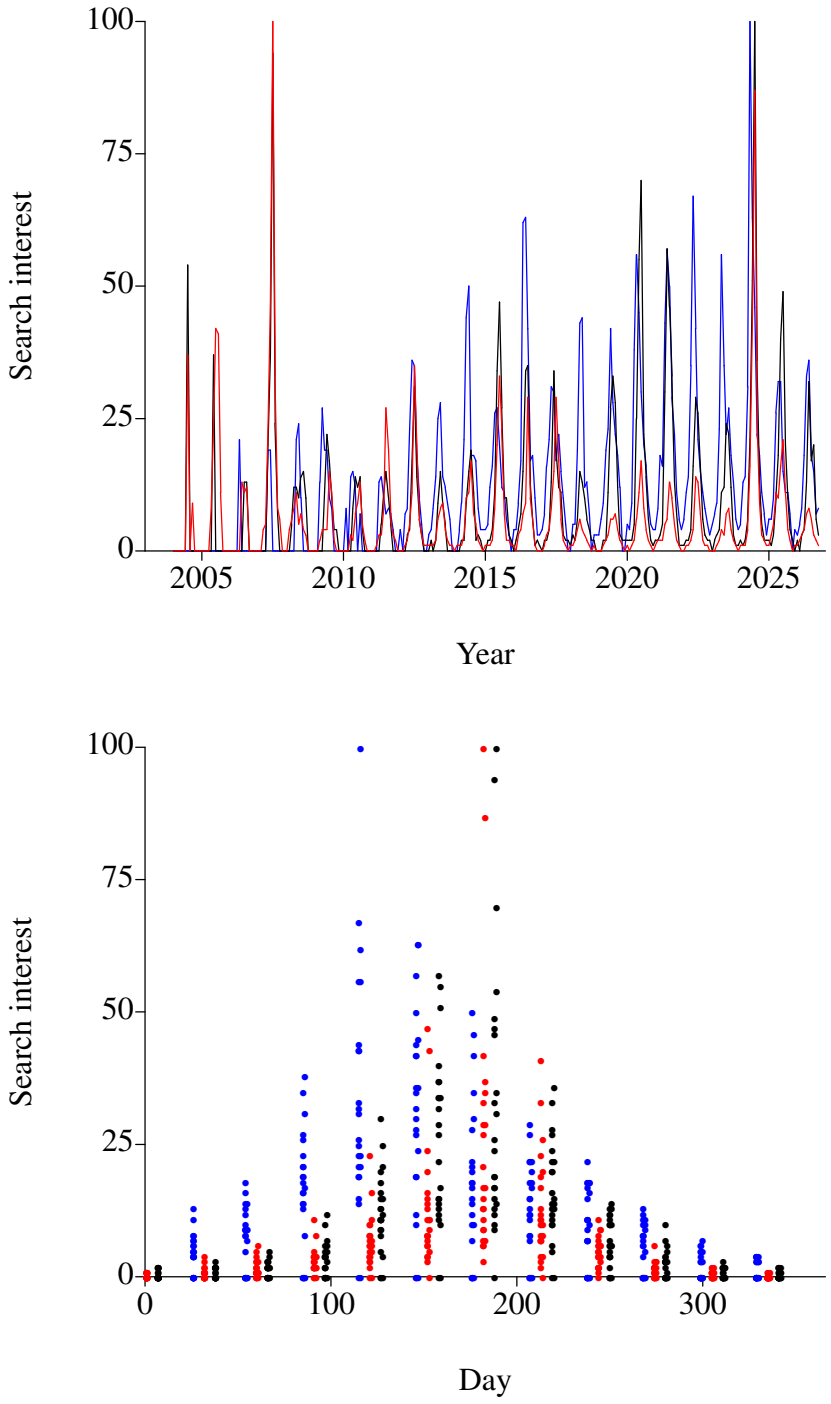



having a cryoprotectant in ones body tissue is similar to have antifreeze in ones coolant

1. The Glucose Accumulation Disparity

Both species trigger glycogen-to-glucose conversion dynamically upon freezing, but their capacity to ramp up glucose levels differs. • Arion rufus and Arion ater complexes demonstrate a more pronounced hyperglycemic response to freezing than Arion vulgaris. • Arion vulgaris accumulates glucose upon freezing, but it does so less efficiently and in lower concentrations compared to its cold-adapted northern relatives. Because of this lower peak concentration, it gets less colligative protection from its sugar reserves.




2. Lower Freezing Survival Limits

Because Arion vulgaris originated from more temperate regions of Europe, its internal machinery is less equipped for prolonged deep freezes. • Survival window: While both species can survive short-term freezing (such as 2 days at a mild -1°C), Arion ater handles prolonged or slightly lower temperatures better. In contrast, Arion vulgaris experiences much higher mortality rates if tissue ice content becomes too high or if temperatures drop to even -2°C for an extended period.

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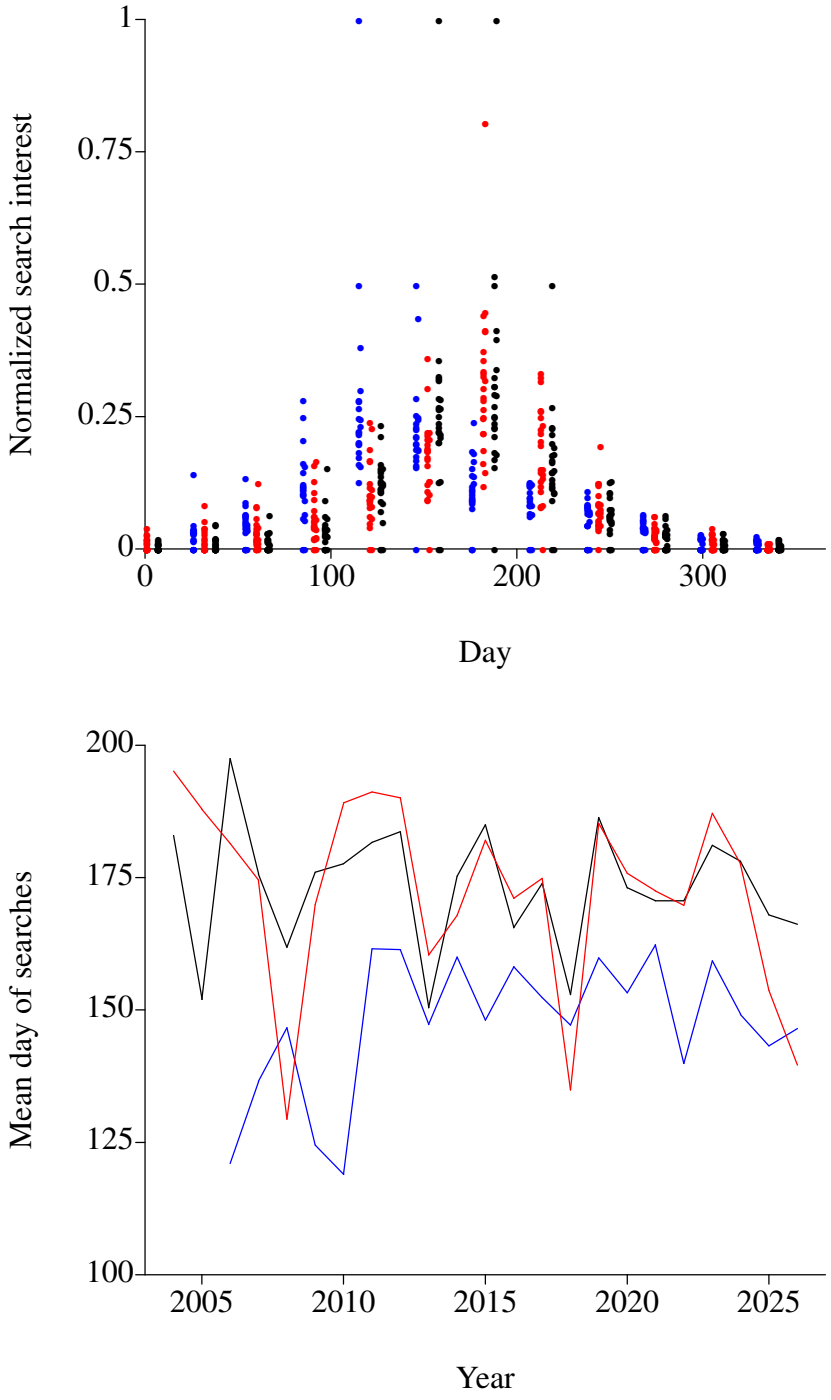


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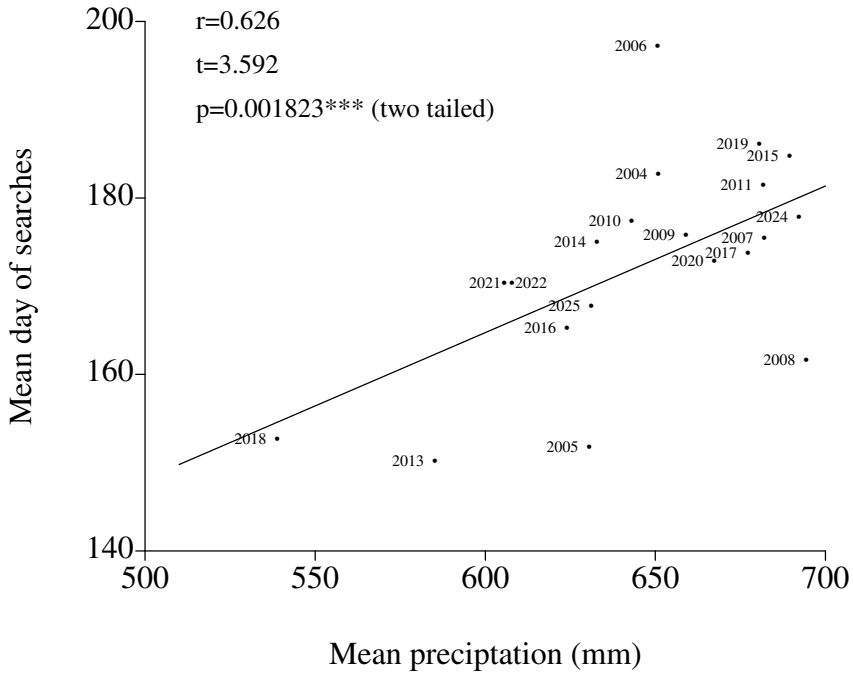


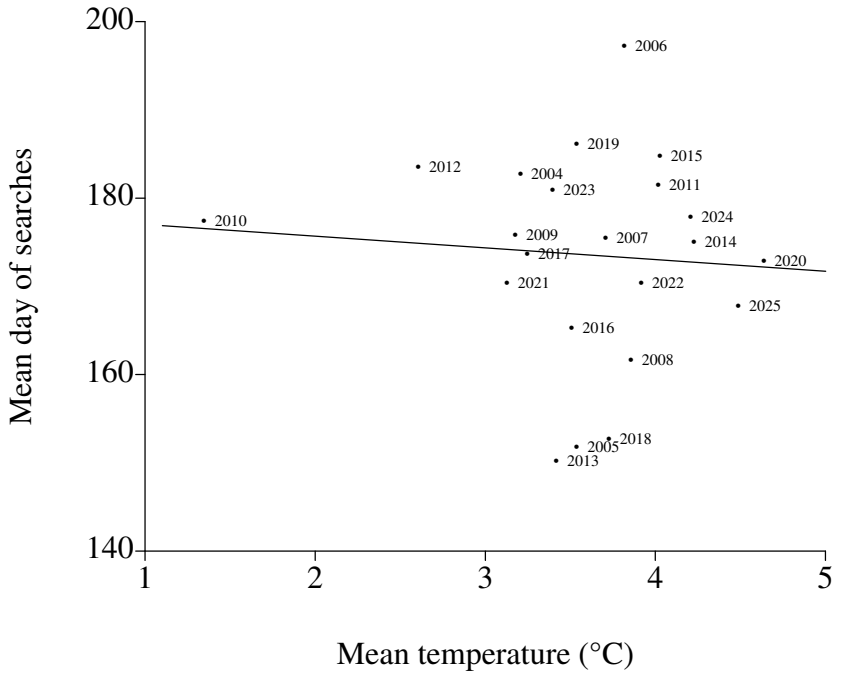


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